VALET: VLM-Assisted LEarning with Targeted queries

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CS-503 Project Proposal

***Abstract*—A major limitation of Reinforcement Learning (RL) is its sample inefficiency: even simple tasks often require thousands of interactions to achieve satisfactory performance. Unlike recent Language Models (LMs), and in particular Vision-Language Models (VLMs), RL agents do not benefit from prior knowledge. In contrast, VLMs possess extensive world knowledge but incur high inference costs.**

In this project, we propose VALET (VLM-Assisted LEarning with Targeted queries), a framework to mitigate RL sample inefficiency by leveraging the knowledge embedded in VLMs. Just as a valet is called upon only when needed rather than accompanying every step, VALET augments a standard RL agent with an additional action that allows it to query a VLM for guidance only when beneficial, enabling more informed action-taking while controlling computational cost.

I. Introduction

Reinforcement Learning (RL) agents are notoriously sample-inefficient, requiring large numbers of interactions to learn effective policies, particularly in sparse-reward settings. Prior work has sought to address this by injecting knowledge from Large Foundation Models (LFMs): through **reward shaping** [1], [2], [3], [4], **Vision-Language-Action (VLA) models** [5], [6], [7], or using foundation models **as direct policies** [8], [9] at every timestep.

These limitations motivate a different approach: rather than replacing the RL policy with a foundation model or using it to engineer rewards, we argue for **selective, on-demand integration** of a VLM into the RL loop. In this work, we propose **VALET** (**V**LM-**A**ssisted **LE**arning with **T**argeted queries). The name reflects the core design philosophy: like a valet who stands ready to assist but does not interfere unbidden, the VLM in VALET is available on demand and serves the agent only when called upon. Inspired by dual-process models of cognition [10], we treat the RL policy as a fast, reactive system, while the VLM acts as a slower, deliberative module. VALET extends the RL agent with an additional action that allows it to query the VLM for guidance, which returns a suggested action based on the current state.

To ensure efficiency, the agent is explicitly penalized for using the VLM, encouraging it to rely on external guidance only when necessary. This setup allows us to study how and when an RL agent benefits from access to external knowledge during learning.

We evaluate the effectiveness of **VALET** by addressing the following research questions:

- Does the proposed framework reduce sample complexity compared to standard RL baselines?
- Does the agent learn to invoke the VLM selectively and efficiently?
- How does the trade-off between performance gains and computational cost evolve?
- How does penalizing VLM usage affect the agent's behavior and performance?
- To what extent does the proposed framework generalize across different tasks and environments?

A. What is the problem you want to solve?

RL agents suffer from sample inefficiency, as they must discover task structure purely through trial and error, with no prior knowledge of the world. VLMs, by contrast, encode rich semantic knowledge but are too expensive to query at every timestep. VALET addresses this tension by augmenting the agent's action space with an explicit "query VLM" action, letting the agent learn through the RL objective itself when consulting the VLM is worth its cost, and when to rely on its own policy.

B. Why is it important that this problem be solved?

Solving this problem would (i) improve **sample efficiency** by bootstrapping exploration with world knowledge, (ii) enable **adaptive use of expensive models** by invoking them only when needed, and (iii) potentially improve **generalization**, as pretrained VLM knowledge may transfer across tasks and environments.

II. Method and Deliveries

A. How do you solve the problem?

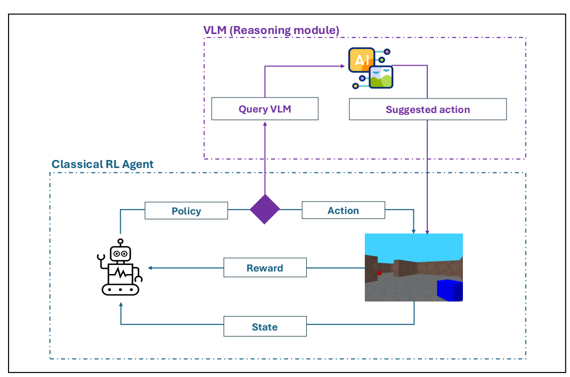

Figure 1. Illustration of our solution

VALET augments the action space of a standard PPO agent with an additional "Query VLM" action. At each timestep, the agent chooses either a regular environment action or to query the VLM, which returns a suggested action given the current visual observation. To discourage unnecessary queries, the agent receives a small negative reward whenever it invokes the VLM. Through training, the agent learns endogenously when external guidance is worth its cost. We use the VLM, prompted with a description of the task and a screenshot of the current state.

B. How will you validate your solution?

We plan to compare VALET against a vanilla PPO baseline across all environments. Our primary quantitative metrics are episodic return and sample efficiency (area under the learning curve). To assess the selective guidance mechanism specifically, we also track VLM query frequency over training, expecting calls to concentrate early and decrease as the policy matures. Qualitatively, we will visualize which states trigger VLM queries to verify that the agent learns meaningful querying behavior rather than querying arbitrarily.

C. Experiments and Timeline

We plan to train VALET across environments of increasing complexity. We will first tackle MiniGrid, a 2D discrete environment with tasks including Goal Reaching and Object Finding. As a proof of concept, we will begin with a fully observable setting to isolate the core mechanism and avoid the additional complexity of a Partially Observable MDP, comparing against a standard PPO baseline. We will then assess whether the policy generalizes to unseen tasks such as FrozenLake. If results are promising and time permits, we would like to push towards more complex environments such as VizDoom, which operates in a continuous action space. We already have preliminary ideas on how to handle this, one direction is to discretize the action space into a finite set of macro-actions that the VLM can select from.

III. Related Work

While VLMs have been used in RL to mitigate sample inefficiency through reward shaping, direct policies, and high-level reasoning in complex domains such as autonomous driving [1], [5], [8], [11], [12], [13], [14], continuous supervision remains computationally prohibitive. To address this, recent frameworks employ selective guidance, triggering VLM interventions based on heuristic measures of uncertainty such as trust metrics [15] or policy entropy.

Most closely related to our work in this latter category is Merler et al. [16], a framework currently under review that we had the opportunity to discuss directly with the authors. They augment a model-free PPO agent with selective VLM supervision triggered by a hand-crafted entropy threshold. While the authors confirmed our problem formulation is complementary to theirs, **VALET** differs in that rather than relying on an exogenous heuristic trigger, VALET integrates the VLM query as an explicit, penalized action within the agent's action space. This creates an endogenous cost-benefit trade-off, allowing the agent to autonomously learn *when* to consult the VLM through the standard RL optimization objective, offering a more general and interpretable formulation.

IV. Discussion

A. Implications and Contribution

We expect **VALET** to demonstrate significantly faster convergence compared to our baselines, effectively reducing sample complexity. Beyond raw performance, we anticipate the emergence of strategic querying behaviors: the agent should learn to prioritize VLM calls during early-stage exploration or in high-uncertainty states, gradually phasing them out as its internal policy matures. The primary contribution of this work is the **endogenization of the guidance mechanism**. While existing literature often relies on hand-crafted heuristics, such as entropy thresholds or fixed schedules, to trigger external help, VALET treats the VLM query as an intrinsic action within the MDP's action space, transforming the problem from heuristic supervision into a learned, cost-aware optimization problem.

B. Feasibility & Risk Assessment

- **Models:** We plan to primarily use Gemma3 (4B) [17], a lightweight open-source VLM that offers a good trade-off between reasoning capability and inference speed. If performance proves insufficient, we will consider scaling to larger models such as Qwen2.5-VL (7B), accepting the associated increase in inference time.
- **Compute:** Following Merler et al. [16], we estimate approximately 2,000 VLM calls over 500k training steps. With Gemma3 (4B) inference taking roughly 1 second per call using vLLM, this amounts to 30–60 minutes of VLM inference time alone, not accounting for PPO optimization. This is a tractable overhead, and we have access to EPFL's Izar cluster (NVIDIA A100 GPUs) to run experiments.
- **Software tools:** All required libraries Gymnasium, MiniGrid, CleanRL, and vLLM are open-source and compatible with the Izar cluster.
- **Planning & Risk:** The project is research-oriented and inherently exploratory, which carries some risk. However, following our discussions with Merler et al., who validated the relevance of our approach, we are confident that the core mechanism is sound. Our PoC on MiniGrid will serve as an early checkpoint: if the VLM query action is not being learned meaningfully, we will pivot to investigate the penalty weighting or action space formulation before moving to more complex environments.

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